

Automated Detection of Keratoconus using EfficientNetB7 : A Deep Learning Approach

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Abstract—A progressive and degenerative eye condition called keratoconus is illustrated by the cornea's thinning and bulging into a cone-like form. The consequence of this deformity is severe visual impairment, including distorted vision, increased sensitivity to light, and potential vision loss if left untreated. Early and precise detection of keratoconus is crucial for improved monitoring and therapeutic outcomes, as timely intervention can prevent disease progression and improve patient outcomes. In this work, we implemented a deep learning approach leveraging the use of EfficientNet-B7 architecture to detect and classify keratoconus, suspect, and normal corneas. EfficientNet-B7, known for its superior performance and scalability, was employed as the backbone of our convolutional neural network (CNN). The dataset used for this study comprised multiple samples, each consisting of seven distinct corneal maps, including topographic, tomographic, and pachymetric representations. These maps provided a holistic view of corneal structure and variations, allowing the model to depict the complex patterns linked to keratoconus. The proposed methodology has achieved a validation accuracy of 97%. This study demonstrates how deep learning can significantly advance eye care, offering precise and early detection of keratoconus by incorporating corneal maps that can improve the quality of life of individuals at risk for keratoconus. This advancement highlights the importance of integrating advanced AI models into clinical practice for the early diagnosis and management of ocular diseases.

Index Terms—Keratoconus(KCN), Cornea, Efficientnetb7, Corneal Maps, Medical Image Processing , Optical Coherence Tomography

I. INTRODUCTION

Keratoconus is a degenerative eye condition in which typically dome-shaped cornea thins and protrudes outward into a cone-like shape. Daily activities like reading and driving become challenging due to the hazy and distorted vision caused by this deformity. The condition typically begins in adolescence and can progress into the mid-30s. It affects approximately 1 in 2,000 individuals globally [1]. Signs of keratoconus include distorted or blurred vision, heightened light sensitivity, and frequent adjustments to prescription

eyeglasses. Soft contact lenses or glasses might be used to handle the early phases. Specialized contact lenses or surgical alternatives like corneal transplants may be required when the situation worsens. [2]. Regular monitoring by an ophthalmologist is crucial for effective management and vision preservation. Uneven astigmatism, often challenging to treat, typically results from corneal thinning and can lead to progressive visual impairment. [3]. For keratoconus detection corneal topography and corneal tomography are the most often used diagnostic techniques [4]. Fig. 1 shows the cornea of Normal eye and keratoconus eye where we can clearly observe bulging of cornea.

Deep learning has revolutionized medical imaging by enhancing the accuracy and efficiency of image analysis, leading to improved patient outcomes. Deep learning techniques may recognize and diagnose illnesses with astonishing precision by utilizing massive datasets. The quality of medical photographs is enhanced using deep learning algorithms, facilitating better visualization and interpretation, which is crucial for accurate diagnosis. EfficientNet model comes under the family of convolutional neural networks (CNNs) developed to offer good performance using limited computational resources. The EfficientNet-B7 model, in particular is the largest variant in this family, offering state-of-the-art accuracy for image classification tasks [5].

The main contributions of this work are summarised as follows.

- To classify ocular condition employing EfficientNetB7, thereby supporting reliable clinical decision-making.
- To incorporate seven corneal maps into your EfficientNetB7 model for detecting normal, keratoconus, and suspect eyes to enhance diagnostic accuracy significantly.
- Lastly, a variety of performance metrics are employed to assess the suggested model's performance.

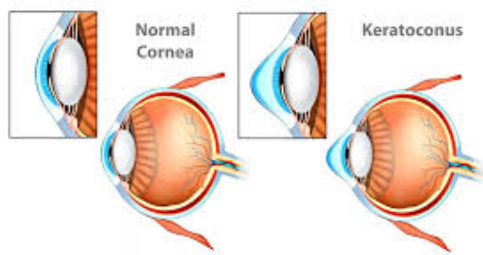


Fig. 1. Normal Eye vs Keratoconus Eye.

II. RELATED WORK

Zorto et al. in 2023 [1] put forth a novel work, which utilizes of machine learning techniques to assess the significance of risk variables in the detection of keratoconus. The study highlights the importance of risk factor information in keratoconus diagnosis and also study indicates that adding risk factor data greatly improves the model's performance. Lavric et al. in 2020 [2] put forth a novel research that explains the development and assessment of a machine learning model method for the early discovery of kcn. OCT-based topographical equipment was used to collect real-world medical data at several Japanese corneal clinics, including corneal topography, elevation, and pachymetry measurement parameters.

Al-Timemy et al. in 2023 [3] developed a deep learning (DL) model based on deep feature fusion for the diagnosis of keratoconus (KCN). The Xception and InceptionResNetV2 DL architectures were applied by the scientists to identify characteristics in three distinct corneal mappings taken from 1371 eyes analyzed at an eye clinic in Egypt. Lavric et al. in 2021 [4] proposed a work to more precisely identify keratoconus (KCN), a machine learning (ML) technique has been developed, put into practice, and validated. An additional machine learning model is developed to identify the most crucial elements in the detection of KCN utilizing corneal data, including elevation, topography, and pachymetry, in order to assess the severity of the condition.

Zhao in 2019 [6] presented a new study that uses photographed images to detect anterior and lateral segments by utilizing the Fusion Features of Anterior and Lateral Segment detection to detect keratoconus using machine learning approaches. To train and assess the model, they use a dataset of 447 parameters from 3164 patient records. Priya in 2022 [7] presented a machine learning model is that successful in determining whether or not a person had keratoconus. Additionally, the developed model was able to determine the disease's severity, which ranges from 1 to 4, with 1 denoting the least severity and 4 the greatest. To give a precise result for the keratoconus detection, the model consider account all of the different scans that are currently performed by medical science

Wang et al. in 2021 [8] suggests a new architecture of the Boundary-Guided Convolutional Neural Network (BG-CNN) is intended to appropriately and automatically separate the

various corneal layers from optical coherence tomography (OCT) pictures. The motivation behind this research stems from the challenges associated with manually segmenting corneal layers, which is time-consuming and prone to variability among readers due to the similar density and texture of adjacent layers. Zhang et al. in 2024 [9] proposed a work that introduces CorNet, for keratoconus (kcn) identification that uses raw biomechanical data from the Corvis ST. This study suggests that CorNet exhibits diagnostic efficacy that is on par with or better than the Corvis ST's current parameters. This suggests that it could improve KC diagnosis in the realm of ophthalmology.

Kanellopoulos et al. in 2024 [10] presented a work that deals with the occurrence of kcn and suspected kcn in Greece is notably higher than previously reported in other places. Specifically the findings indicate that approximately 11.08 of patients had topographic keratoconus, Furthermore 25.12% were identified as having suspect keratoconus. This suggests that one in ten Greeks may have keratoconus, with many cases remaining undiagnosed even by the time of cataract surgery. Sandhiya et al. in 2024 [11] proposed the work which focuses about how machine learning techniques can be used for Preliminary spotting of keratoconus. The work highlights the importance of early identification to improve patient quality of life and proposes a collaborative approach between medical professionals and AI experts to enhance diagnostic methods.

Muhsin et al. in year 2024 [13] proposed a novel approach to detect keratoconus, a degenerative corneal disease, Employing machine learning models. To improve the precision and consistency of keratoconus diagnosis, the authors suggest applying a number of machine learning approaches, including as Random Forest, Optimized k-Nearest Neighbors (k-NN), Decision Trees (DT), Support Vector Machines (SVM), and Logistic Regression (LR). Kallel et al. in 2024 [14] presented a deep learning-based method for corneal topographic map-based early keratoconus detection, a critical tool for diagnosing various corneal conditions. By employing CNNs, the research aims to enhance the classification accuracy of these maps, which reveal intricate details about corneal shape and irregularities.

Nayeem, et al. in 2024 [15] put forth a study concludes that pre-trained Convolutional Neural Networks (CNNs) exhibit varying degrees of effectiveness in detecting keratoconus, normal, and suspect cases. MobileNetV2 emerged as the most accurate model, showcasing excellent precision and recall with minimal misclassifications across all categories. Table I presents the comparative analyses of some of the research work with the proposed work.

III. MATERIALS AND PROPOSED METHODOLOGY

A. Dataset

Three experts with corneal training carried out the eye classification. Each of the seven corneal maps in the train validation set is 150 Keratoconus, 150 Normal, and 123

TABLE I
SUMMARY OF STUDIES ON KCN DETECTION USING DEEP LEARNING

Study	KCN Categories	Device/Imaging System	Evaluation Strategy	Data Size / Maps	Model Used	Reported Accuracy
Kamiya et al. [21]	Normal, Four KCN Grades	Tomey/CASIA	Five-fold Cross-Validation	543 cases / 6 maps	ResNet-18	99%
Kuto et al. [22]	Normal vs. KCN	Tomey TMS-4 Corneal Topographer	Training, Testing, Subclinical Assessment	354 cases / 1 map	VGG16, InceptionV3	93.1%
Lavric & Valentin [23]	Normal vs. KCN	Simulated Maps	Training, Validation, Testing	SyntEyes, KTCmodels maps	KeratoDetect	99.3%
Al-Timemy et al. [24]	Normal vs. KCN	OCULUS Pentacam	Training, Validation, Testing	534 cases / 4 maps	EDTL, AlexNet	98.3%
Al-Timemy, Ali H., et al. [15]	Normal, KCN, Suspected KCN	OCULUS Pentacam	Training, Validation, Independent Testing	692 eyes / 7 maps	EfficientNet-B0, SVM	98% (2-class), 81.6% (3-class)
Present Study	Normal, KCN, Suspected KCN	OCULUS Pentacam	Training, Validation, Independent Testing	4011 cases / 7 maps	EfficientNet-B7	97% (3-class)

suspected images. Each test set contains seven corneal maps. 50 Suspect images, 50 KCN, and 50 NOR [?]. The doctors were told to assign a grade of normal, suspected KCN, or KCN to each eye. More precisely, eyes were classified as suspected KCN if they had an asymmetrical bowtie pattern or peculiar, regional steepening of their corneal surface. If the central corneal thickness was less than 500 microns, An eye was classified as likely keratoconus (KCN) if the oblique cylinder exceeded 1.50 diopters or the keratometric curvature was greater than 47.00 diopters. Corneal photos taken with various Pentacam equipment and settings are included in the training dataset (the map's color scale levels change from those of the previous subset).

B. Proposed Methodology

The proposed methodology is implemented using following steps. Data preparation, model intialisation, compilation, training and evaluation. Fig. 2 depicts the proposed methodology. EfficientNet is a family of convolutional neural networks (CNNs) developed by Google AI, designed to achieve high performance with fewer computational resources. The EfficientNet-B7 model, in particular is the largest variant in this family, for image classification tasks. EfficientNet balances the network's depth, width, and resolution using a special scaling technique known as compound scaling. EfficientNet-B7 was pre-trained on ImageNet. In order to fine-tune for this particular activity, the top layers were eliminated. To allow for fine-tuning, the model was configured to be trainable. Within the EfficientNet network family, there are variations among the quantity of MBConv blocks. From EfficientNetB0 to EfficientNetB7, depth, width, resolution, and model size all continue to grow, and accuracy also becomes better. Fig. 3 shows the architecture of the efficientnetb7 model.

1) *Preprocessing*: The following procedures were used in order to get the data ready for model training and assessment:

- **Resizing**: Every corneal map image was downsized to fit the EfficientNet-B7 model's required input dimensions, such as 224x224 pixels.

- **Normalization**: Pixel values were scaled to a range of [0, 1] to ensure uniform input and facilitate model convergence during training.
- **Data Augmentation**: On the training set, data augmentation approaches were used to reduce overfitting and enhance model generalizability. They were random rotations, Flips. Flips for both horizontal and vertical Adjustments for contrast and brightness

2) Feature extraction:

- **Initial Convolution and Pooling**: The preprocessed image passes through an initial convolutional layer followed by a pooling layer, which captures basic features like edges and textures.
- **MBConv Blocks**: The core of EfficientNet-B7 consists of multiple Mobile Inverted Bottleneck Convolution (MBConv) blocks. These blocks efficiently extract hierarchical features, capturing complex patterns and structures within the image.

3) Classification:

- **Global Average Pooling**: Feature maps from the final MBConv block are aggregated into a single vector by computing the average value of each feature map, reducing spatial dimensions.
- **Fully Connected Layer**: This vector is fed into a dense layer that maps the extracted features to a set of class probabilities.
- **Softmax Activation**: The softmax function in the output layer produces a probability distribution over all possible classes, which helps with the final classification choice. The figure 3 shows the proposed methodology for classification of keratoconus, normal and suspected eyes.

4) *MBConv Block*: Each MBConv block consists of depth-wise separable convolutions and a squeeze-and-excitation module, which together enhance feature representation while maintaining computational efficiency. As the image progresses through these blocks, it undergoes transformations that capture hierarchical features at various levels of abstraction. The Mobile Inverted Bottleneck Convolution (MBConv), is

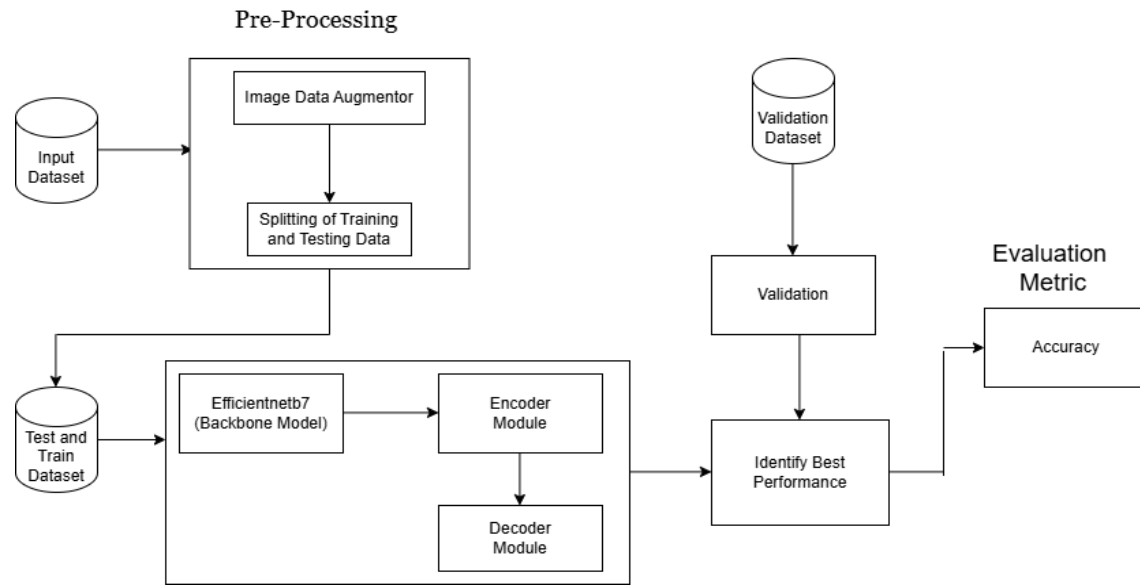


Fig. 2. Proposed Methodology

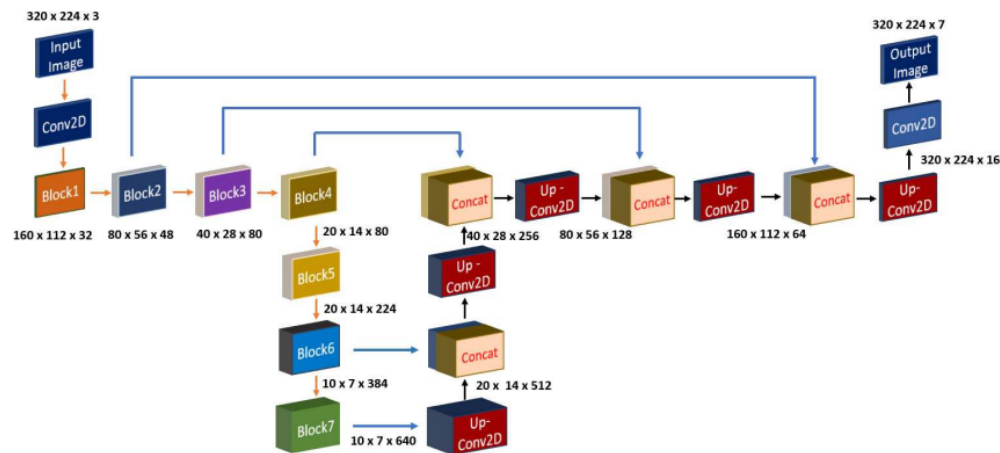


Fig. 3. Architecture of Efficientnetb7

a fundamental building block in modern convolutional neural network (CNN) architecture. Key Components of the MBConv Block are

- Expansion Layer: First, a 1×1 convolutional layer is applied to the input tensor, increasing the number of channels. This expansion allows the network to capture a broader range of features.
- Depthwise Separable Convolution: Following the expansion, a depthwise separable convolution is applied. It involves two steps: (a) Depthwise Convolution: Each input channel is convolved with its own set of filters, preserving spatial information while reducing compu-

tational complexity. (b) Pointwise Convolution: A 1×1 convolution is applied across all channels to combine the outputs, enabling the network to learn complex feature interactions

- Squeeze and Excitation (SE) Block (Optional): Some implementations contain a SE block that dynamically adjusts the importance of each channel responses by capturing relationships among feature channels. This mechanism enhances the representational capacity of the network.
- Projection Layer: The enlarged feature map is projected back to a lower-dimensional space after features are

processed through a 1x1 convolutional layer, reducing the number of channels and computational load.

- **Residual Connection:** A skip connection adds the input tensor to the output of the projection layer, facilitating gradient flow during training and aiding in the convergence of deep networks. Fig. 4 shows working of MBConv block.

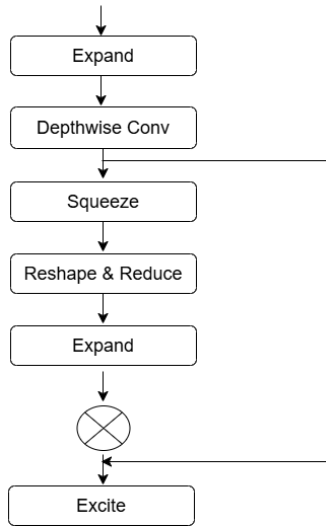


Fig. 4. MBConv: Basic building block of EfficientNet.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this study, the EfficientNet-B7 architecture was employed for the identification of keratoconus, normal and suspected kcn using a dataset comprising corneal maps categorized into three classes Normal (NOR), Keratoconus (KCN), and Suspect. Each sample consisted of seven corneal maps. Dataset being splitted into training, validation, and test sets. Fig. 5 shows the variation between training and validation loss, with the y-axis plotted with loss and the x-axis plotted with the number of epochs, indicating that the loss is decreasing across the epochs. Fig. 6 shows the validation and training accuracy of the model with x-axis being plotted with epochs and y-axis with accuracy which depicts that curve keeps increasing as expected.

Fig. 7 shows the confusion matrix for the proposed model correctly identified 247 cases of keratoconus, 52 cases of normal and 307 cases of suspect. Table II gives the comparative analysis of EfficientNetB7 with EfficientNetB4 which shows EfficientNetB4 gives an accuracy of 93.5%. The proposed model outperformed EfficientNetB4 with an accuracy of 97%, demonstrating its efficiency in distinguishing between these categories. These results demonstrate how deep learning methods might improve ophthalmological diagnoses, especially for diseases like keratoconus where early diagnosis is essential.

V. CONCLUSION AND FUTURE WORK

This current study provides an innovative approach for detecting keratoconus using deep learning techniques,

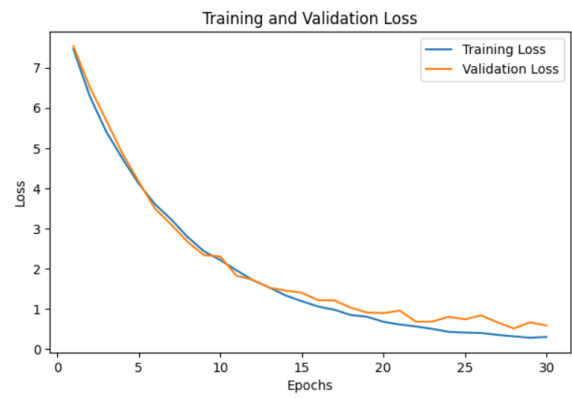


Fig. 5. Training and Validation Loss.

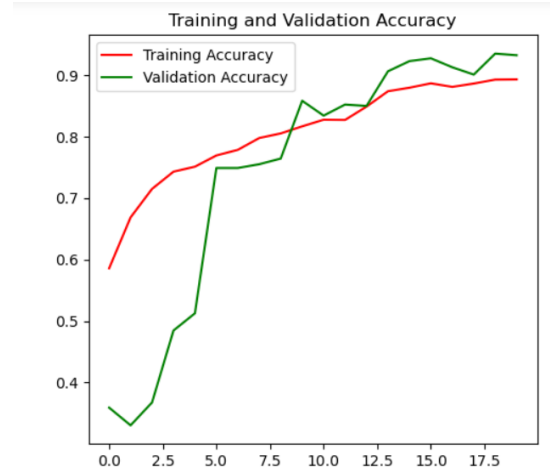


Fig. 6. Training and Validation Accuracy.

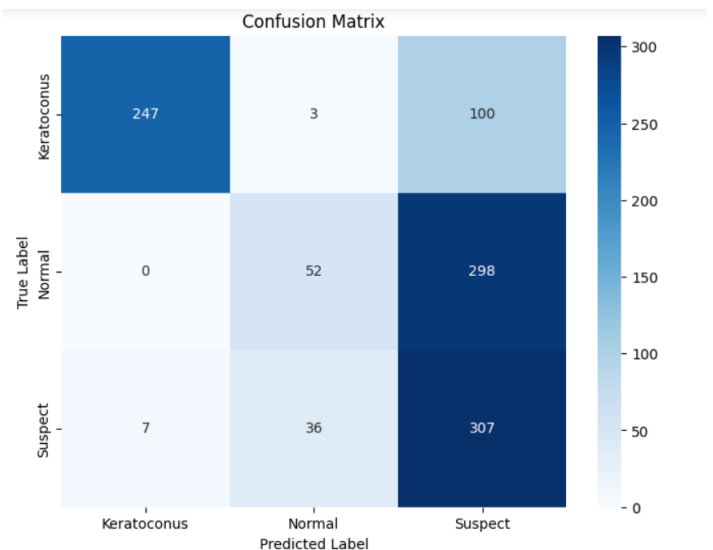


Fig. 7. Confusion Matrix

specifically by employing EfficientNet-B7 architecture.

TABLE II
COMPARATIVE ANALYSIS OF PROPOSED MODEL

Model	Class	Precision	Recall	F1 Score	Accuracy
EfficientNetB4	Keratoconus	0.89	0.51	0.73	93.5
	Normal	0.93	0.11	0.20	
	suspect	0.91	0.79	0.83	
EfficientNetB7	keratoconus	0.98	0.71	0.83	97
	Normal	1.00	0.14	0.25	
	suspect	0.97	0.89	0.93	

The study provides the significance of early detection of keratoconus, a progressive eye disorder, to improve patient outcomes and prevent vision loss. By leveraging the capabilities of EfficientNet-B7, known for its optimal balance of accuracy and computational efficiency, the proposed model achieved a remarkable validation accuracy of 97%. The high validation accuracy underscores the efficacy of the model in distinguishing between normal eyes, suspect cases, and confirmed keratoconus cases, demonstrating its potential as a reliable diagnostic tool. This study adds to the expanding corpus of knowledge in AI and medical imaging by offering a robust framework for tackling complex ophthalmological conditions.

In future research, expanding the dataset with more diverse and high-quality corneal scans can enhance the model's generalization and robustness. Classification performance might be further enhanced by investigating different deep learning architectures, such as Vision Transformers or hybrid models. Additionally, for practical deployment, the model must be deployed in real-time clinical situations and its performance validated by ophthalmologists.

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